Тема: Настройка протоколов динамической маршрутизации RIP v2 и OSPF

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

1) Для заданной на схеме schema-lab5 сети, состоящей из управляемых коммутаторов, маршрутизаторов и персональных компьютеров

выполнить планирование и документирование адресного пространства и назначить статические адреса всем устройствам.

nb! Каждое соединение маршрутизатора с маршрутизатором - это отдельная сеть.

2) Настроить протокол динамической маршрутизации RIP v2 для области, указанной на схеме schema-lab5.

3) Настроить протокол динамической маршрутизации OSPF для зон 0, 1, 2. Зону 1 настроить как полностью (nb!) тупиковую.

4) Настроить редистрибуцию маршрутов между протоколами RIP v2 и OSPF.

5) Проверить работоспособность маршрутизации, выполнив ping VPC "все между всеми" (nb!: в обе стороны).

6) Перехватить в wireshark сообщения протоколов RIP v2 и OSPF, идентифицировать их тип и содержание.

7) Сохранить в отдельные файлы с префиксом rt_ и именем маршрутизатора таблицы маршрутизации всех маршрутизаторов.

8) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.

1.

Адресный план:

RIP v2 область

Соединение	Сеть	Устройство	Интерфейс	IP
PC1 — R4	192.168.41.0/24	R4	Fa0/0	192.168.41.1
PC1 — R4	192.168.41.0/24	PC1	Eth0	192.168.41.10
R4 — R5	172.16.45.0/30	R4	Fa1/0	172.16.45.1
R4 — R5	172.16.45.0/30	R5	Fa1/0	172.16.45.2
PC2 — R5	192.168.52.0/24	R5	Fa0/0	192.168.52.1
PC2 — R5	192.168.52.0/24	PC2	Eth0	192.168.52.10
R5 — R1	172.16.15.0/30	R5	Fa2/0	172.16.15.1
R5 — R1	172.16.15.0/30	R1	Fa0/0	172.16.15.2

OSPF Area 0

Устройство	Интерфейс	IP
R1	Fa1/0	10.0.123.1/24
R2	Fa0/0	10.0.123.2/24
R3	Fa0/0	10.0.123.3/24

сеть 10.0.123.0/24

OSPF Area 2

Соединение	Сеть	Устройство	Интерфейс	IP
R2 — R6	10.0.26.0/30	R2	Fa1/0	10.0.26.1
R2 — R6	10.0.26.0/30	R6	Fa0/0	10.0.26.2
R3 — R7	10.0.37.0/30	R3	Fa1/0	10.0.37.1
R3 — R7	10.0.37.0/30	R7	Fa0/0	10.0.37.2
R6 — R7	10.0.67.0/30	R6	Fa2/0	10.0.67.1
R6 — R7	10.0.67.0/30	R7	Fa2/0	10.0.67.2
PC3 — R6	192.168.63.0/24	R6	Fa1/0	192.168.63.1

PC3 — R6	192.168.63.0/24	PC3	Eth0	192.168.63.10
PC4 — R7	192.168.74.0/24	R7	Fa1/0	192.168.74.1
PC4 — R7	192.168.74.0/24	PC4	Eth0	192.168.74.10

OSPF Area 1, полностью тупиковая

Соединение	Сеть	Устройство	Интерфейс	IP
R3 — R8	10.0.38.0/30	R3	Fa2/0	10.0.38.1
R3 — R8	10.0.38.0/30	R8	Fa0/0	10.0.38.2
PC5 — R8	192.168.85.0/24	R8	Fa1/0	192.168.85.1
PC5 — R8	192.168.85.0/24	PC5	Eth0	192.168.85.10

Настройка статической маршрутизации:

R1

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#description to R5
R1(config-if)#ip address 172.16.15.2 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:06:45.027: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
*Mar  1 00:06:46.027: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R1(config)#interface FastEthernet1/0
R1(config-if)#description to Layer2Switch-1 Area 0
R1(config-if)#ip address 10.0.123.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:06:58.759: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:06:59.759: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R1(config)#end
R1#
```

```
*Mar  1 00:07:02.159: %SYS-5-CONFIG_I: Configured from console by
console
R1#wr
Building configuration...
[OK]
```

R2

```
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#hostname R2
R2(config)#interface FastEthernet0/0
R2(config-if)#description to Layer2Switch-1 Area 0
R2(config-if)#ip address 10.0.123.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar  1 00:06:41.507: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
*Mar  1 00:06:42.507: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R2(config)#interface FastEthernet1/0
R2(config-if)#description to R6
R2(config-if)#ip address 10.0.26.1 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Mar  1 00:06:54.539: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:06:55.539: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R2(config)#end
R2#
*Mar  1 00:06:57.631: %SYS-5-CONFIG_I: Configured from console by
console
R2#wr
Building configuration...
[OK]
```

R3

```
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#hostname R3
R3(config)#interface FastEthernet0/0
R3(config-if)#description to Layer2Switch-1 Area 0
R3(config-if)#ip address 10.0.123.3 255.255.255.0
R3(config-if)#no shutdown
```

```
R3(config-if)#exit
R3(config)#
*Mar  1 00:07:05.447: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
*Mar  1 00:07:06.447: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R3(config)#interface FastEthernet1/0
R3(config-if)#description to R7
R3(config-if)#ip address 10.0.37.1 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar  1 00:07:19.303: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:07:20.303: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R3(config)#interface FastEthernet2/0
R3(config-if)#description to R8
R3(config-if)#ip address 10.0.38.1 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Mar  1 00:07:34.023: %LINK-3-UPDOWN: Interface FastEthernet2/0,
changed state to up
*Mar  1 00:07:35.023: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet2/0, changed state to up
R3(config)#end
R3#
*Mar  1 00:07:37.415: %SYS-5-CONFIG_I: Configured from console by
console
R3#wr
Building configuration...
[OK]
```

R4

```
R4#enable
R4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#hostname R4
R4(config)#interface FastEthernet0/0
R4(config-if)#description to PC1
R4(config-if)#ip address 192.168.41.1 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar  1 00:07:32.875: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
```

```
*Mar  1 00:07:33.875: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R4(config)#interface FastEthernet1/0
R4(config-if)#description to R5
R4(config-if)#ip address 172.16.45.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Mar  1 00:07:50.303: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:07:51.303: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R4(config)#end
R4#
*Mar  1 00:07:54.155: %SYS-5-CONFIG_I: Configured from console by
console
R4#wr
Building configuration...
[OK]
```

R5

```
R5#enable
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#hostname R5
R5(config)#interface FastEthernet0/0
R5(config-if)#description to PC2
R5(config-if)#ip address 192.168.52.1 255.255.255.0
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar  1 00:08:12.843: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
*Mar  1 00:08:13.843: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R5(config)#interface FastEthernet1/0
R5(config-if)#description to R4
R5(config-if)#ip address 172.16.45.2 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar  1 00:08:27.627: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:08:28.627: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R5(config)#interface FastEthernet2/0
R5(config-if)#description to R1
R5(config-if)#
```

```
R5(config-if)#ip address 172.16.15.1 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Mar  1 00:08:56.347: %LINK-3-UPDOWN: Interface FastEthernet2/0,
changed state to up
*Mar  1 00:08:57.347: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet2/0, changed state to up
R5(config)#end
R5#
*Mar  1 00:08:59.975: %SYS-5-CONFIG_I: Configured from console by
console
R5#wr
Building configuration...
[OK]
```

R6

```
R6#enable
R6#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#hostname R6
R6(config)#interface FastEthernet0/0
R6(config-if)#
R6(config-if)#
R6(config-if)#description to R2
R6(config-if)#ip address 10.0.26.2 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
*Mar  1 00:09:22.239: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
*Mar  1 00:09:23.239: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R6(config)#interface FastEthernet1/0
R6(config-if)#description to PC3
R6(config-if)#ip address 192.168.63.1 255.255.255.0
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
*Mar  1 00:09:36.959: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:09:37.959: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R6(config)#interface FastEthernet2/0
R6(config-if)#description to R7
R6(config-if)#ip address 10.0.67.1 255.255.255.252
R6(config-if)#no shutdown
```

```
R6(config-if)#exit
R6(config)#
*Mar  1 00:09:51.647: %LINK-3-UPDOWN: Interface FastEthernet2/0,
changed state to up
*Mar  1 00:09:52.647: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet2/0, changed state to up
R6(config)#end
R6#
*Mar  1 00:09:54.959: %SYS-5-CONFIG_I: Configured from console by
console
R6#wr
Building configuration...
[OK]
```

R7

```
R7#enable
R7#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#hostname R7
R7(config)#interface FastEthernet0/0
R7(config-if)#description to R3
R7(config-if)#ip address 10.0.37.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#
*Mar  1 00:10:07.903: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
*Mar  1 00:10:08.903: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R7(config-if)#exit
R7(config)#interface FastEthernet1/0
R7(config-if)#description to PC4
R7(config-if)#ip address 192.168.74.1 255.255.255.0
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Mar  1 00:10:28.943: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:10:29.943: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R7(config)#interface FastEthernet2/0
R7(config-if)#description to R6
R7(config-if)#ip address 10.0.67.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Mar  1 00:10:43.759: %LINK-3-UPDOWN: Interface FastEthernet2/0,
changed state to up
```

```
*Mar  1 00:10:44.759: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet2/0, changed state to up
R7(config)#end
R7#
*Mar  1 00:10:48.759: %SYS-5-CONFIG_I: Configured from console by
console
R7#wr
Building configuration...
[OK]
```

R8

```
R8#enable
R8#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#hostname R8
R8(config)#interface FastEthernet0/0
R8(config-if)#description to R3
R8(config-if)#ip address 10.0.38.2 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
*Mar  1 00:10:48.763: %LINK-3-UPDOWN: Interface FastEthernet0/0,
changed state to up
*Mar  1 00:10:49.763: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet0/0, changed state to up
R8(config)#interface FastEthernet1/0
R8(config-if)#description to PC5
R8(config-if)#ip address 192.168.85.1 255.255.255.0
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
*Mar  1 00:11:03.979: %LINK-3-UPDOWN: Interface FastEthernet1/0,
changed state to up
*Mar  1 00:11:04.979: %LINEPROTO-5-UPDOWN: Line protocol on
Interface FastEthernet1/0, changed state to up
R8(config)#end
R8#
*Mar  1 00:11:07.751: %SYS-5-CONFIG_I: Configured from console by
console
R8#wr
Building configuration...
[OK]
```

Настройка компьютеров:

PC1

```
PC1> ip 192.168.41.10/24 192.168.41.1
Checking for duplicate address...
PC1 : 192.168.41.10 255.255.255.0 gateway 192.168.41.1
```

```
PC1> save
Saving startup configuration to startup.vpc
.  done
```

PC2

```
PC2> ip 192.168.52.10/24 192.168.52.1
Checking for duplicate address...
PC2 : 192.168.52.10 255.255.255.0 gateway 192.168.52.1

PC2> save
Saving startup configuration to startup.vpc
.  done
```

PC3

```
PC3> ip 192.168.63.10/24 192.168.63.1
Checking for duplicate address...
PC3 : 192.168.63.10 255.255.255.0 gateway 192.168.63.1

PC3> save
Saving startup configuration to startup.vpc
.  done
```

PC4

```
PC4> ip 192.168.74.10/24 192.168.74.1
Checking for duplicate address...
PC4 : 192.168.74.10 255.255.255.0 gateway 192.168.74.1

PC4> save
Saving startup configuration to startup.vpc
.  done
```

PC5

```
PC5> ip 192.168.85.10/24 192.168.85.1
Checking for duplicate address...
PC5 : 192.168.85.10 255.255.255.0 gateway 192.168.85.1

PC5> save
Saving startup configuration to startup.vpc
.  done
```

2. Настройка RIP v2

На R1 в RIP включаем только сеть между R1 и R5:

R1

```
R1#enable
```

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#network 172.16.0.0
R1(config-router)#end
R1#
*Mar  1 00:41:55.259: %SYS-5-CONFIG_I: Configured from console by
console
R1#wr
Building configuration...
[OK]
```

На R4 в RIP включаем сеть к PC1 и сеть к R5:

R4

```
R4#enable
R4#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 172.16.0.0
R4(config-router)#network 192.168.41.0
R4(config-router)#end
R4#wr
Building configuration...

*Mar  1 00:38:02.535: %SYS-5-CONFIG_I: Configured from console by
console[OK]
```

R5

На R5 в RIP включаем сеть к PC2, сеть к R4 и сеть к R1:

```
R5#enable
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#no auto-summary
R5(config-router)#network 172.16.0.0
R5(config-router)#network 192.168.52.0
R5(config-router)#end
R5#
*Mar  1 00:38:35.263: %SYS-5-CONFIG_I: Configured from console by
console
R5#wr
Building configuration...
[OK]
```

Проверка соединения на компьютерах

PC1

```
PC1> ping 192.168.52.10

192.168.52.10 icmp_seq=1 timeout
84 bytes from 192.168.52.10 icmp_seq=2 ttl=62 time=25.042 ms
84 bytes from 192.168.52.10 icmp_seq=3 ttl=62 time=28.825 ms
84 bytes from 192.168.52.10 icmp_seq=4 ttl=62 time=31.054 ms
84 bytes from 192.168.52.10 icmp_seq=5 ttl=62 time=28.286 ms
```

PC2

```
PC2> ping 192.168.41.10

84 bytes from 192.168.41.10 icmp_seq=1 ttl=62 time=37.425 ms
84 bytes from 192.168.41.10 icmp_seq=2 ttl=62 time=28.913 ms
84 bytes from 192.168.41.10 icmp_seq=3 ttl=62 time=28.150 ms
84 bytes from 192.168.41.10 icmp_seq=4 ttl=62 time=29.179 ms
84 bytes from 192.168.41.10 icmp_seq=5 ttl=62 time=30.287 ms
```

3. Настройка OSPF

R1

R1 находится в OSPF Area 0.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 10.0.123.0 0.0.0.255 area 0
R1(config-router)#end
R1#
*Mar  1 00:46:47.035: %SYS-5-CONFIG_I: Configured from console by
console
R1#wr
Building configuration...
[OK]
```

R2

R2 находится в Area 0 и Area 2.

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#router-id 2.2.2.2
R2(config-router)#network 10.0.123.0 0.0.0.255 area 0
R2(config-router)#network 10.0.26.0 0.0.0.3 area 2
R2(config-router)#end
R2#
```

```
*Mar  1 00:45:28.991: %SYS-5-CONFIG_I: Configured from console by
console
R2#wr
Building configuration...
[OK]
R2#
*Mar  1 00:45:32.299: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done
```

R3

R3 находится в Area 0, Area 2 и Area 1.

Он является ABR для полностью тупиковой зоны 1.

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#router-id 3.3.3.3
R3(config-router)#network 10.0.123.0 0.0.0.255 area 0
R3(config-router)#network 10.0.37.0 0.0.0.3 area 2
R3(config-router)#
*Mar  1 00:45:22.243: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Done
*Mar  1 00:45:22.251: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R3(config-router)#network 10.0.38.0 0.0.0.3 area 1
R3(config-router)#area 1 stub no-summary
R3(config-router)#end
R3#wr
*Mar  1 00:45:37.223: %SYS-5-CONFIG_I: Configured from console by
console
R3#wr
Building configuration...
[OK]
```

```
area 1 stub no-summary
```

ставится именно на R3, потому что R3 — пограничный маршрутизатор зоны, то есть ABR.

R6

R6 находится в Area 2.

```
R6#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#router ospf 1
R6(config-router)#router-id 6.6.6.6
R6(config-router)#network 10.0.26.0 0.0.0.3 area 2
R6(config-router)#network 10.0.67.0 0.0.0.3 area 2
*Mar  1 00:42:44.871: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R6(config-router)#network 10.0.67.0 0.0.0.3 area 2
R6(config-router)#
```

```
R6(config-router)#network 192.168.63.0 0.0.0.255 area 2
R6(config-router)#end
R6#
*Mar  1 00:43:03.111: %SYS-5-CONFIG_I: Configured from console by
console
R6#wr
Building configuration...
[OK]
```

R7

R7 находится в Area 2.

```
R7#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.37.0 0.0.0.3 area 2
R7(config-router)#network 10.0.67.0 0.0.0.3 area 2
*Mar  1 00:42:42.611: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on
FastEthernet0/0 from LOADING to FULL, Loading Done
R7(config-router)#network 10.0.67.0 0.0.0.3 area 2
R7(config-router)#network 192.168.74.0 0.0.0.255 area 2
*Mar  1 00:42:49.163: %OSPF-5-ADJCHG: Process 1, Nbr 6.6.6.6 on
FastEthernet2/0 from LOADING to FULL, Loading Done
R7(config-router)#network 192.168.74.0 0.0.0.255 area 2
R7(config-router)#end
R7#
*Mar  1 00:42:55.543: %SYS-5-CONFIG_I: Configured from console by
console
R7#wr
Building configuration...
[OK]
```

R8

R8 находится внутри Area 1.

```
R8#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#router ospf 1
R8(config-router)#router-id 8.8.8.8
R8(config-router)#network 10.0.38.0 0.0.0.3 area 1
R8(config-router)#network 192.168.85.0 0.0.0.255 area 1
R8(config-router)#area 1 stub
R8(config-router)#end
R8#wr
Building configuration...
[OK]
```

На внутреннем маршрутизаторе зоны 1 пишется:

area 1 stub

А на ABR R3 пишется:
area 1 stub no-summary
Именно так зона 1 становится полностью тупиковой.
`show ip route`

```
R8#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.0.38.1 to network 0.0.0.0

     10.0.0.0/30 is subnetted, 1 subnets
C       10.0.38.0 is directly connected, FastEthernet0/0
C    192.168.85.0/24 is directly connected, FastEthernet1/0
O*IA 0.0.0.0/0 [110/2] via 10.0.38.1, 00:01:30, FastEthernet0/0
```

R8 не получает подробные межзональные маршруты, а получает только default-route из Area 1 наружу.

4. Настройка редистрибуции RIP v2 ↔ OSPF

R1

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#no auto-summary
R1(config-router)#redistribute ospf 1 metric 2
R1(config-router)#exit
R1(config)#router ospf 1
R1(config-router)#
R1(config-router)#redistribute rip subnets
R1(config-router)#exit
R1(config)#end
R1#QP:
*Mar  1 00:55:53.875: %SYS-5-CONFIG_I: Configured from console by
console
R1#wr
Building configuration...
[OK]
```

`redistribute ospf 1 metric 2`
означает: маршруты из OSPF process 1 будут передаваться в RIP.
Метрика 2 нужна обязательно, потому что RIP использует hop count.
`redistribute rip subnets`
означает: маршруты из RIP будут передаваться в OSPF.
subnets обязательно, потому что без него OSPF может не перераспределить подсети с масками /24 и /30 корректно.

Проверка редистрибуции:

show ip route

R4

```
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

R    192.168.63.0/24 [120/3] via 172.16.45.2, 00:00:24, FastEthernet1/0
R    192.168.74.0/24 [120/3] via 172.16.45.2, 00:00:24, FastEthernet1/0
     172.16.0.0/30 is subnetted, 2 subnets
C       172.16.45.0 is directly connected, FastEthernet1/0
R       172.16.15.0 [120/1] via 172.16.45.2, 00:00:24, FastEthernet1/0
C    192.168.41.0/24 is directly connected, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
R       10.0.26.0/30 [120/3] via 172.16.45.2, 00:00:24, FastEthernet1/0
R       10.0.38.0/30 [120/3] via 172.16.45.2, 00:00:26, FastEthernet1/0
R       10.0.37.0/30 [120/3] via 172.16.45.2, 00:00:26, FastEthernet1/0
R       10.0.67.0/30 [120/3] via 172.16.45.2, 00:00:26, FastEthernet1/0
R       10.0.123.0/24 [120/3] via 172.16.45.2, 00:00:26, FastEthernet1/0
R    192.168.52.0/24 [120/1] via 172.16.45.2, 00:00:26, FastEthernet1/0
R    192.168.85.0/24 [120/3] via 172.16.45.2, 00:00:04, FastEthernet1/0
```

R5

```
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

R    192.168.63.0/24 [120/2] via 172.16.15.2, 00:00:19, FastEthernet2/0
R    192.168.74.0/24 [120/2] via 172.16.15.2, 00:00:19, FastEthernet2/0
     172.16.0.0/30 is subnetted, 2 subnets
C       172.16.45.0 is directly connected, FastEthernet1/0
C       172.16.15.0 is directly connected, FastEthernet2/0
R    192.168.41.0/24 [120/1] via 172.16.45.1, 00:00:11, FastEthernet1/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
R       10.0.26.0/30 [120/2] via 172.16.15.2, 00:00:19, FastEthernet2/0
R       10.0.38.0/30 [120/2] via 172.16.15.2, 00:00:21, FastEthernet2/0
R       10.0.37.0/30 [120/2] via 172.16.15.2, 00:00:21, FastEthernet2/0
R       10.0.67.0/30 [120/2] via 172.16.15.2, 00:00:21, FastEthernet2/0
R       10.0.123.0/24 [120/2] via 172.16.15.2, 00:00:21, FastEthernet2/0
C    192.168.52.0/24 is directly connected, FastEthernet0/0
R    192.168.85.0/24 [120/2] via 172.16.15.2, 00:00:23, FastEthernet2/0
```

Появились маршруты R, значит, что OSPF-маршруты были переданы в RIP

R2,R3,R6,R7,R8

```
R2#show ip route ospf
O    192.168.63.0/24 [110/2] via 10.0.26.2, 00:08:22, FastEthernet1/0
O    192.168.74.0/24 [110/3] via 10.0.26.2, 00:08:22, FastEthernet1/0
     172.16.0.0/30 is subnetted, 2 subnets
O E2    172.16.45.0 [110/20] via 10.0.123.1, 00:03:34, FastEthernet0/0
O E2    172.16.15.0 [110/20] via 10.0.123.1, 00:03:34, FastEthernet0/0
O E2 192.168.41.0/24 [110/20] via 10.0.123.1, 00:03:34, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O IA    10.0.38.0/30 [110/2] via 10.0.123.3, 00:03:34, FastEthernet0/0
O       10.0.37.0/30 [110/3] via 10.0.26.2, 00:08:22, FastEthernet1/0
O       10.0.67.0/30 [110/2] via 10.0.26.2, 00:08:22, FastEthernet1/0
O E2 192.168.52.0/24 [110/20] via 10.0.123.1, 00:03:34, FastEthernet0/0
O IA 192.168.85.0/24 [110/3] via 10.0.123.3, 00:03:34, FastEthernet0/0
```

```
R3#show ip route ospf
O    192.168.63.0/24 [110/3] via 10.0.37.2, 00:08:25, FastEthernet1/0
O    192.168.74.0/24 [110/2] via 10.0.37.2, 00:08:25, FastEthernet1/0
     172.16.0.0/30 is subnetted, 2 subnets
O E2    172.16.45.0 [110/20] via 10.0.123.1, 00:03:30, FastEthernet0/0
O E2    172.16.15.0 [110/20] via 10.0.123.1, 00:03:30, FastEthernet0/0
O E2 192.168.41.0/24 [110/20] via 10.0.123.1, 00:03:30, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O       10.0.26.0/30 [110/3] via 10.0.37.2, 00:08:25, FastEthernet1/0
O       10.0.67.0/30 [110/2] via 10.0.37.2, 00:08:25, FastEthernet1/0
O E2 192.168.52.0/24 [110/20] via 10.0.123.1, 00:03:30, FastEthernet0/0
O    192.168.85.0/24 [110/2] via 10.0.38.2, 00:07:31, FastEthernet2/0
```

```
R6#show ip route ospf
O    192.168.74.0/24 [110/2] via 10.0.67.2, 00:08:34, FastEthernet2/0
     172.16.0.0/30 is subnetted, 2 subnets
O E2    172.16.45.0 [110/20] via 10.0.26.1, 00:03:40, FastEthernet0/0
O E2    172.16.15.0 [110/20] via 10.0.26.1, 00:03:40, FastEthernet0/0
O E2 192.168.41.0/24 [110/20] via 10.0.26.1, 00:03:40, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O IA    10.0.38.0/30 [110/3] via 10.0.67.2, 00:08:34, FastEthernet2/0
                     [110/3] via 10.0.26.1, 00:08:34, FastEthernet0/0
O       10.0.37.0/30 [110/2] via 10.0.67.2, 00:08:34, FastEthernet2/0
O IA    10.0.123.0/24 [110/2] via 10.0.26.1, 00:08:34, FastEthernet0/0
O E2 192.168.52.0/24 [110/20] via 10.0.26.1, 00:03:40, FastEthernet0/0
O IA 192.168.85.0/24 [110/4] via 10.0.67.2, 00:07:37, FastEthernet2/0
                     [110/4] via 10.0.26.1, 00:07:37, FastEthernet0/0
```

```
R7#show ip route ospf
O    192.168.63.0/24 [110/2] via 10.0.67.1, 00:08:36, FastEthernet2/0
     172.16.0.0/30 is subnetted, 2 subnets
O E2    172.16.45.0 [110/20] via 10.0.37.1, 00:03:44, FastEthernet0/0
O E2    172.16.15.0 [110/20] via 10.0.37.1, 00:03:44, FastEthernet0/0
O E2 192.168.41.0/24 [110/20] via 10.0.37.1, 00:03:44, FastEthernet0/0
     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O       10.0.26.0/30 [110/2] via 10.0.67.1, 00:08:36, FastEthernet2/0
O IA    10.0.38.0/30 [110/2] via 10.0.37.1, 00:08:36, FastEthernet0/0
O IA    10.0.123.0/24 [110/2] via 10.0.37.1, 00:08:36, FastEthernet0/0
O E2 192.168.52.0/24 [110/20] via 10.0.37.1, 00:03:44, FastEthernet0/0
O IA 192.168.85.0/24 [110/3] via 10.0.37.1, 00:07:41, FastEthernet0/0
```

```
R8#show ip route ospf
O*IA 0.0.0.0/0 [110/2] via 10.0.38.1, 00:07:41, FastEthernet0/0
```

Внешние OSPF маршруты к RIP сетям с пометкой O E2.
На R8 подробных внешних маршрутов O E2 нет.
На R1
`show ip protocols`

```
R1#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 27 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip, ospf 1
  Default version control: send version 2, receive version 2
    Interface             Send  Recv  Triggered RIP  Key-chain
    FastEthernet0/0       2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    172.16.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.15.1          120      00:00:23
  Distance: (default is 120)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  It is an autonomous system boundary router
```

видно, что RIP перераспределяет OSPF, а OSPF перераспределяет RIP.

5. Проверка ping между всеми VPC

PC1 PC2

```
PC1> ping 192.168.52.10

84 bytes from 192.168.52.10 icmp_seq=1 ttl=62 time=39.426 ms
84 bytes from 192.168.52.10 icmp_seq=2 ttl=62 time=28.020 ms
84 bytes from 192.168.52.10 icmp_seq=3 ttl=62 time=26.198 ms
84 bytes from 192.168.52.10 icmp_seq=4 ttl=62 time=26.651 ms
84 bytes from 192.168.52.10 icmp_seq=5 ttl=62 time=27.757 ms

PC1> ping 192.168.63.10

192.168.63.10 icmp_seq=1 timeout
84 bytes from 192.168.63.10 icmp_seq=2 ttl=59 time=70.571 ms
84 bytes from 192.168.63.10 icmp_seq=3 ttl=59 time=53.007 ms
84 bytes from 192.168.63.10 icmp_seq=4 ttl=59 time=60.458 ms
84 bytes from 192.168.63.10 icmp_seq=5 ttl=59 time=51.532 ms

PC1> ping 192.168.74.10

192.168.74.10 icmp_seq=1 timeout
84 bytes from 192.168.74.10 icmp_seq=2 ttl=59 time=59.344 ms
84 bytes from 192.168.74.10 icmp_seq=3 ttl=59 time=67.776 ms
84 bytes from 192.168.74.10 icmp_seq=4 ttl=59 time=58.179 ms
84 bytes from 192.168.74.10 icmp_seq=5 ttl=59 time=67.610 ms

PC1> ping 192.168.85.10

192.168.85.10 icmp_seq=1 timeout
84 bytes from 192.168.85.10 icmp_seq=2 ttl=59 time=60.290 ms
84 bytes from 192.168.85.10 icmp_seq=3 ttl=59 time=56.622 ms
84 bytes from 192.168.85.10 icmp_seq=4 ttl=59 time=57.365 ms
84 bytes from 192.168.85.10 icmp_seq=5 ttl=59 time=59.696 ms
```

```
PC2> ping 192.168.41.10

84 bytes from 192.168.41.10 icmp_seq=1 ttl=62 time=21.120 ms
84 bytes from 192.168.41.10 icmp_seq=2 ttl=62 time=27.262 ms
84 bytes from 192.168.41.10 icmp_seq=3 ttl=62 time=29.511 ms
84 bytes from 192.168.41.10 icmp_seq=4 ttl=62 time=28.818 ms
84 bytes from 192.168.41.10 icmp_seq=5 ttl=62 time=28.118 ms

PC2> ping 192.168.63.10

84 bytes from 192.168.63.10 icmp_seq=1 ttl=60 time=47.607 ms
84 bytes from 192.168.63.10 icmp_seq=2 ttl=60 time=57.218 ms
84 bytes from 192.168.63.10 icmp_seq=3 ttl=60 time=46.942 ms
84 bytes from 192.168.63.10 icmp_seq=4 ttl=60 time=48.241 ms
84 bytes from 192.168.63.10 icmp_seq=5 ttl=60 time=47.185 ms

PC2> ping 192.168.74.10

84 bytes from 192.168.74.10 icmp_seq=1 ttl=60 time=58.691 ms
84 bytes from 192.168.74.10 icmp_seq=2 ttl=60 time=47.303 ms
84 bytes from 192.168.74.10 icmp_seq=3 ttl=60 time=49.192 ms
84 bytes from 192.168.74.10 icmp_seq=4 ttl=60 time=47.867 ms
84 bytes from 192.168.74.10 icmp_seq=5 ttl=60 time=47.609 ms

PC2> ping 192.168.85.10

84 bytes from 192.168.85.10 icmp_seq=1 ttl=60 time=47.205 ms
84 bytes from 192.168.85.10 icmp_seq=2 ttl=60 time=46.465 ms
84 bytes from 192.168.85.10 icmp_seq=3 ttl=60 time=47.260 ms
84 bytes from 192.168.85.10 icmp_seq=4 ttl=60 time=46.575 ms
84 bytes from 192.168.85.10 icmp_seq=5 ttl=60 time=48.915 ms
```

PC3 PC4

```
PC3> ping 192.168.41.10

84 bytes from 192.168.41.10 icmp_seq=1 ttl=59 time=77.451 ms
84 bytes from 192.168.41.10 icmp_seq=2 ttl=59 time=58.977 ms
84 bytes from 192.168.41.10 icmp_seq=3 ttl=59 time=69.811 ms
84 bytes from 192.168.41.10 icmp_seq=4 ttl=59 time=67.547 ms
84 bytes from 192.168.41.10 icmp_seq=5 ttl=59 time=58.597 ms

PC3> ping 192.168.52.10

84 bytes from 192.168.52.10 icmp_seq=1 ttl=60 time=66.312 ms
84 bytes from 192.168.52.10 icmp_seq=2 ttl=60 time=47.371 ms
84 bytes from 192.168.52.10 icmp_seq=3 ttl=60 time=47.653 ms
84 bytes from 192.168.52.10 icmp_seq=4 ttl=60 time=49.891 ms
84 bytes from 192.168.52.10 icmp_seq=5 ttl=60 time=49.432 ms

PC3> ping 192.168.74.10

84 bytes from 192.168.74.10 icmp_seq=1 ttl=62 time=21.265 ms
84 bytes from 192.168.74.10 icmp_seq=2 ttl=62 time=20.628 ms
84 bytes from 192.168.74.10 icmp_seq=3 ttl=62 time=27.580 ms
84 bytes from 192.168.74.10 icmp_seq=4 ttl=62 time=26.936 ms
84 bytes from 192.168.74.10 icmp_seq=5 ttl=62 time=26.893 ms

PC3> ping 192.168.85.10

84 bytes from 192.168.85.10 icmp_seq=1 ttl=60 time=45.764 ms
84 bytes from 192.168.85.10 icmp_seq=2 ttl=60 time=51.392 ms
84 bytes from 192.168.85.10 icmp_seq=3 ttl=60 time=48.851 ms
84 bytes from 192.168.85.10 icmp_seq=4 ttl=60 time=46.852 ms
84 bytes from 192.168.85.10 icmp_seq=5 ttl=60 time=48.352 ms
```

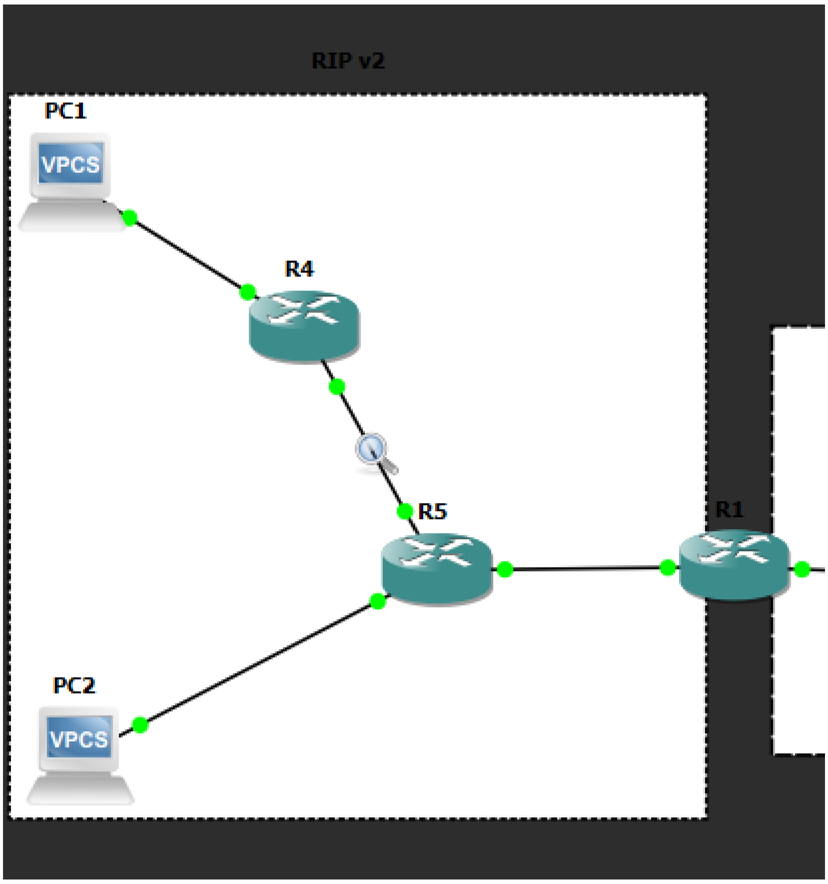
```
PC4> ping 192.168.41.10

84 bytes from 192.168.41.10 icmp_seq=1 ttl=59 time=57.588 ms
84 bytes from 192.168.41.10 icmp_seq=2 ttl=59 time=66.243 ms
84 bytes from 192.168.41.10 icmp_seq=3 ttl=59 time=68.998 ms
84 bytes from 192.168.41.10 icmp_seq=4 ttl=59 time=79.562 ms
84 bytes from 192.168.41.10 icmp_seq=5 ttl=59 time=57.273 ms

PC4> ping 192.168.52.10

84 bytes from 192.168.52.10 icmp_seq=1 ttl=60 time=46.800 ms
84 bytes from 192.168.52.10 icmp_seq=2 ttl=60 time=47.549 ms
84 bytes from 192.168.52.10 icmp_seq=3 ttl=60 time=47.530 ms
84 bytes from 192.168.52.10 icmp_seq=4 ttl=60 time=40.710 ms
84 bytes from 192.168.52.10 icmp_seq=5 ttl=60 time=59.827 ms

PC4> ping 192.168.63.10

84 bytes from 192.168.63.10 icmp_seq=1 ttl=62 time=28.427 ms
84 bytes from 192.168.63.10 icmp_seq=2 ttl=62 time=27.217 ms
84 bytes from 192.168.63.10 icmp_seq=3 ttl=62 time=22.924 ms
84 bytes from 192.168.63.10 icmp_seq=4 ttl=62 time=23.007 ms
84 bytes from 192.168.63.10 icmp_seq=5 ttl=62 time=27.908 ms

PC4> ping 192.168.85.10

84 bytes from 192.168.85.10 icmp_seq=1 ttl=61 time=36.870 ms
84 bytes from 192.168.85.10 icmp_seq=2 ttl=61 time=38.450 ms
84 bytes from 192.168.85.10 icmp_seq=3 ttl=61 time=39.336 ms
84 bytes from 192.168.85.10 icmp_seq=4 ttl=61 time=39.608 ms
84 bytes from 192.168.85.10 icmp_seq=5 ttl=61 time=36.634 ms
```

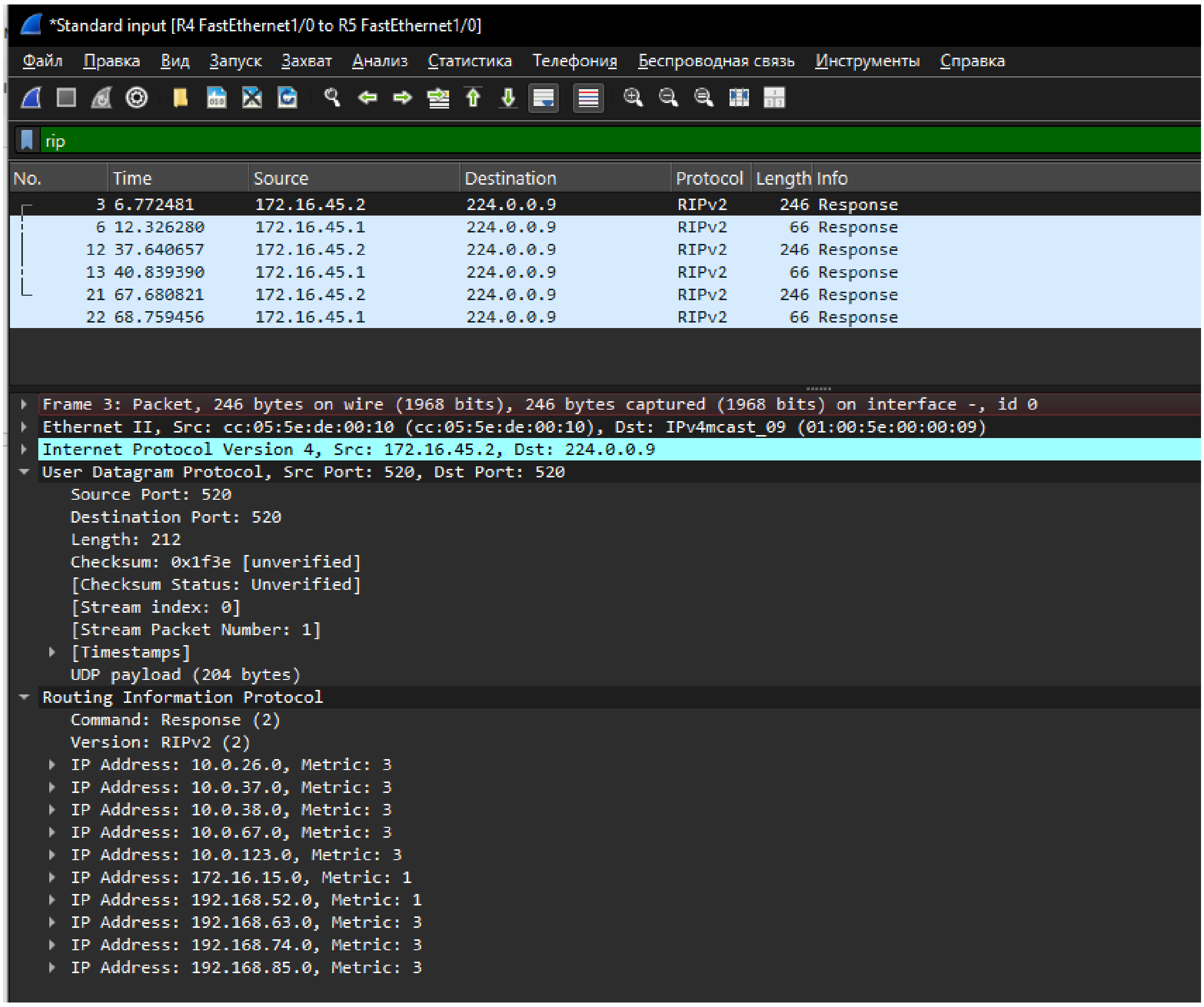
PC5

```
PC5> ping 192.168.41.10

84 bytes from 192.168.41.10 icmp_seq=1 ttl=59 time=69.501 ms
84 bytes from 192.168.41.10 icmp_seq=2 ttl=59 time=89.858 ms
84 bytes from 192.168.41.10 icmp_seq=3 ttl=59 time=67.540 ms
84 bytes from 192.168.41.10 icmp_seq=4 ttl=59 time=69.381 ms
84 bytes from 192.168.41.10 icmp_seq=5 ttl=59 time=58.096 ms

PC5> ping 192.168.52.10

84 bytes from 192.168.52.10 icmp_seq=1 ttl=60 time=57.411 ms
84 bytes from 192.168.52.10 icmp_seq=2 ttl=60 time=69.589 ms
84 bytes from 192.168.52.10 icmp_seq=3 ttl=60 time=51.015 ms
84 bytes from 192.168.52.10 icmp_seq=4 ttl=60 time=58.394 ms
84 bytes from 192.168.52.10 icmp_seq=5 ttl=60 time=47.252 ms

PC5> ping 192.168.63.10

84 bytes from 192.168.63.10 icmp_seq=1 ttl=60 time=52.585 ms
84 bytes from 192.168.63.10 icmp_seq=2 ttl=60 time=59.209 ms
84 bytes from 192.168.63.10 icmp_seq=3 ttl=60 time=41.534 ms
84 bytes from 192.168.63.10 icmp_seq=4 ttl=60 time=56.746 ms
84 bytes from 192.168.63.10 icmp_seq=5 ttl=60 time=48.169 ms

PC5> ping 192.168.74.10

84 bytes from 192.168.74.10 icmp_seq=1 ttl=61 time=47.339 ms
84 bytes from 192.168.74.10 icmp_seq=2 ttl=61 time=39.045 ms
84 bytes from 192.168.74.10 icmp_seq=3 ttl=61 time=39.010 ms
84 bytes from 192.168.74.10 icmp_seq=4 ttl=61 time=37.999 ms
84 bytes from 192.168.74.10 icmp_seq=5 ttl=61 time=49.745 ms
```

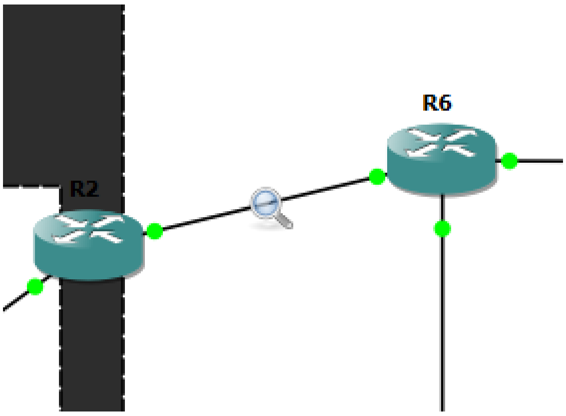
Соединение есть.

6. Для анализа взял R4 FastEthernet1/0 --- R5 FastEthernet1/0



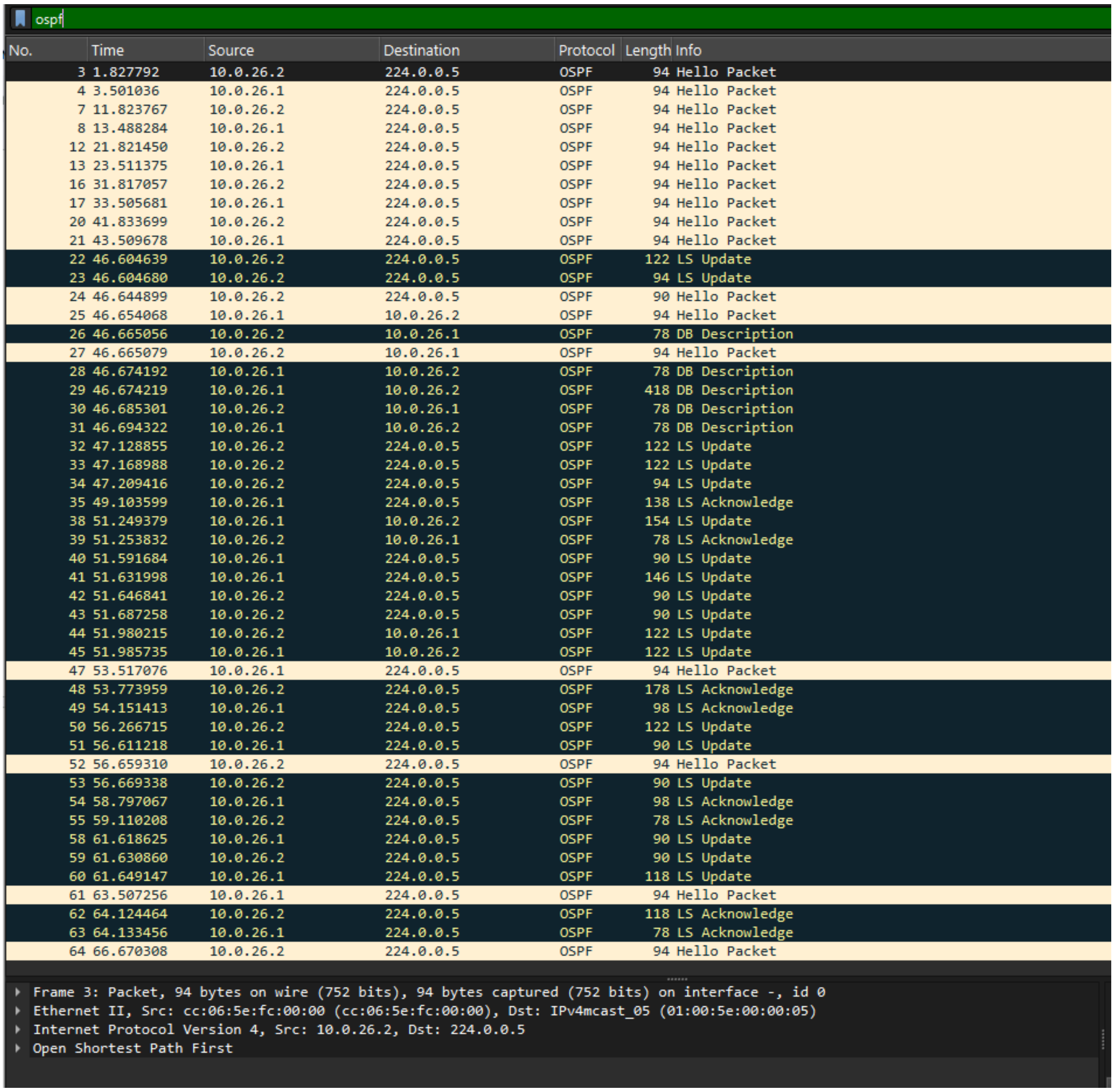


Source IP: 172.16.45.2
Destination IP: 224.0.0.9
Protocol: RIPv2
Command: Response
Version: RIPv2
UDP Source Port: 520
UDP Destination Port: 520
Адрес источника 172.16.45.2 принадлежит интерфейсу маршрутизатора R5, подключенному к R4. Адрес назначения 224.0.0.9 является multicast-адресом, который используется RIP v2 для рассылки маршрутных обновлений.
Внутри RIP-пакета содержатся маршруты, они показывают, какие сети известны маршрутизатору R5 и передаются соседнему маршрутизатору R4.
Маршруты 172.16.15.0 и 192.168.52.0 имеют метрику 1, так как находятся ближе к R5. Остальные сети имеют метрику 3, так как они были получены через редистрибуцию из OSPF и находятся дальше по топологии.
Таким образом, перехват RIP v2 подтвердил, что маршрутизатор R5 отправляет маршрутные обновления соседу R4 с использованием UDP-порта 520 и multicast-адреса 224.0.0.9
OSPF:
R2 FastEthernet1/0 --- R6 FastEthernet0/0



clear ip ospf process

```
R6#clear ip ospf process
Reset ALL OSPF processes? [no]: yes
R6#
*Mar  1 01:04:57.955: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on FastEthernet2/0
from FULL to DOWN, Neighbor Down: Interface down or detached
*Mar  1 01:04:57.959: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0
from FULL to DOWN, Neighbor Down: Interface down or detached
*Mar  1 01:04:58.019: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on FastEthernet2/0
from LOADING to FULL, Loading Done
*Mar  1 01:04:58.023: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet0/0
from LOADING to FULL, Loading Done
R6#
```



ospf

No.	Time	Source	Destination	Protocol	Length	Info
3	1.827792	10.0.26.2	224.0.0.5	OSPF	94	Hello Packet
4	3.501036	10.0.26.1	224.0.0.5	OSPF	94	Hello Packet
7	11.823767	10.0.26.2	224.0.0.5	OSPF	94	Hello Packet
8	13.488284	10.0.26.1	224.0.0.5	OSPF	94	Hello Packet
12	21.821450	10.0.26.2	224.0.0.5	OSPF	94	Hello Packet
13	23.511375	10.0.26.1	224.0.0.5	OSPF	94	Hello Packet
16	31.817057	10.0.26.2	224.0.0.5	OSPF	94	Hello Packet
17	33.505681	10.0.26.1	224.0.0.5	OSPF	94	Hello Packet
20	41.833699	10.0.26.2	224.0.0.5	OSPF	94	Hello Packet
21	43.509678	10.0.26.1	224.0.0.5	OSPF	94	Hello Packet
22	46.604639	10.0.26.2	224.0.0.5	OSPF	122	LS Update
23	46.604680	10.0.26.2	224.0.0.5	OSPF	94	LS Update
24	46.644899	10.0.26.2	224.0.0.5	OSPF	90	Hello Packet
25	46.654068	10.0.26.1	10.0.26.2	OSPF	94	Hello Packet
26	46.665056	10.0.26.2	10.0.26.1	OSPF	78	DB Description
27	46.665079	10.0.26.2	10.0.26.1	OSPF	94	Hello Packet
28	46.674192	10.0.26.1	10.0.26.2	OSPF	78	DB Description
29	46.674219	10.0.26.1	10.0.26.2	OSPF	418	DB Description
30	46.685301	10.0.26.2	10.0.26.1	OSPF	78	DB Description
31	46.694322	10.0.26.1	10.0.26.2	OSPF	78	DB Description
32	47.128855	10.0.26.2	224.0.0.5	OSPF	122	LS Update
33	47.168988	10.0.26.2	224.0.0.5	OSPF	122	LS Update
34	47.209416	10.0.26.2	224.0.0.5	OSPF	94	LS Update
35	49.103599	10.0.26.1	224.0.0.5	OSPF	138	LS Acknowledge
38	51.249379	10.0.26.1	10.0.26.2	OSPF	154	LS Update
39	51.253832	10.0.26.2	10.0.26.1	OSPF	78	LS Acknowledge
40	51.591684	10.0.26.1	224.0.0.5	OSPF	90	LS Update
41	51.631998	10.0.26.1	224.0.0.5	OSPF	146	LS Update
42	51.646841	10.0.26.2	224.0.0.5	OSPF	90	LS Update
43	51.687258	10.0.26.2	224.0.0.5	OSPF	90	LS Update
44	51.980215	10.0.26.2	10.0.26.1	OSPF	122	LS Update
45	51.985735	10.0.26.1	10.0.26.2	OSPF	122	LS Update
47	53.517076	10.0.26.1	224.0.0.5	OSPF	94	Hello Packet
48	53.773959	10.0.26.2	224.0.0.5	OSPF	178	LS Acknowledge
49	54.151413	10.0.26.1	224.0.0.5	OSPF	98	LS Acknowledge
50	56.266715	10.0.26.2	224.0.0.5	OSPF	122	LS Update
51	56.611218	10.0.26.1	224.0.0.5	OSPF	90	LS Update
52	56.659310	10.0.26.2	224.0.0.5	OSPF	94	Hello Packet
53	56.669338	10.0.26.2	224.0.0.5	OSPF	90	LS Update
54	58.797067	10.0.26.1	224.0.0.5	OSPF	98	LS Acknowledge
55	59.110208	10.0.26.2	224.0.0.5	OSPF	78	LS Acknowledge
58	61.618625	10.0.26.1	224.0.0.5	OSPF	90	LS Update
59	61.630860	10.0.26.2	224.0.0.5	OSPF	90	LS Update
60	61.649147	10.0.26.1	224.0.0.5	OSPF	118	LS Update
61	63.507256	10.0.26.1	224.0.0.5	OSPF	94	Hello Packet
62	64.124464	10.0.26.2	224.0.0.5	OSPF	118	LS Acknowledge
63	64.133456	10.0.26.1	224.0.0.5	OSPF	78	LS Acknowledge
64	66.670308	10.0.26.2	224.0.0.5	OSPF	94	Hello Packet

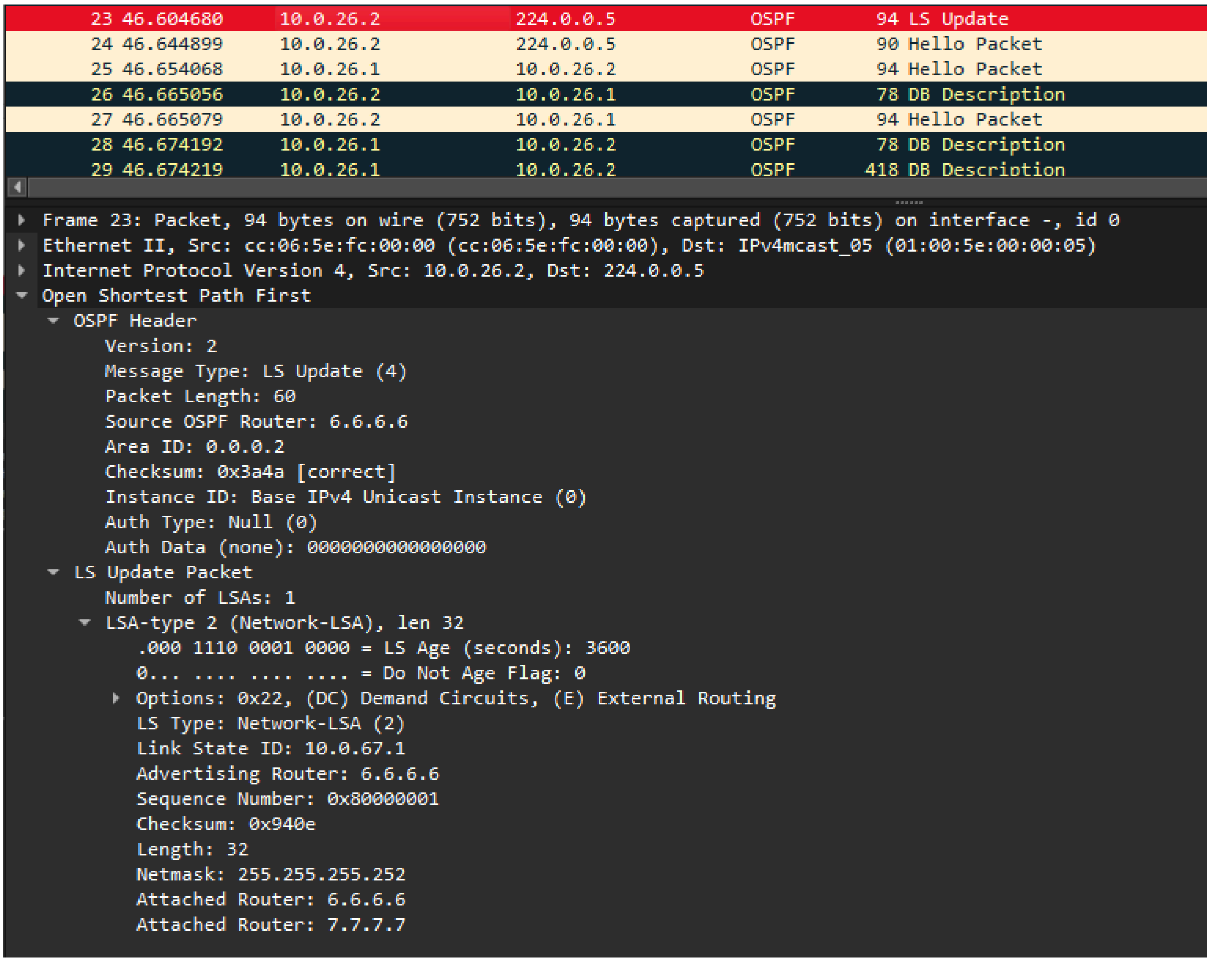
Frame 3: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface -, id 0
Ethernet II, Src: cc:06:5e:fc:00:00 (cc:06:5e:fc:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
Internet Protocol Version 4, Src: 10.0.26.2, Dst: 224.0.0.5
Open Shortest Path First

В захвате были обнаружены
Hello Packet
DB Description
LS Update
LS Acknowledge

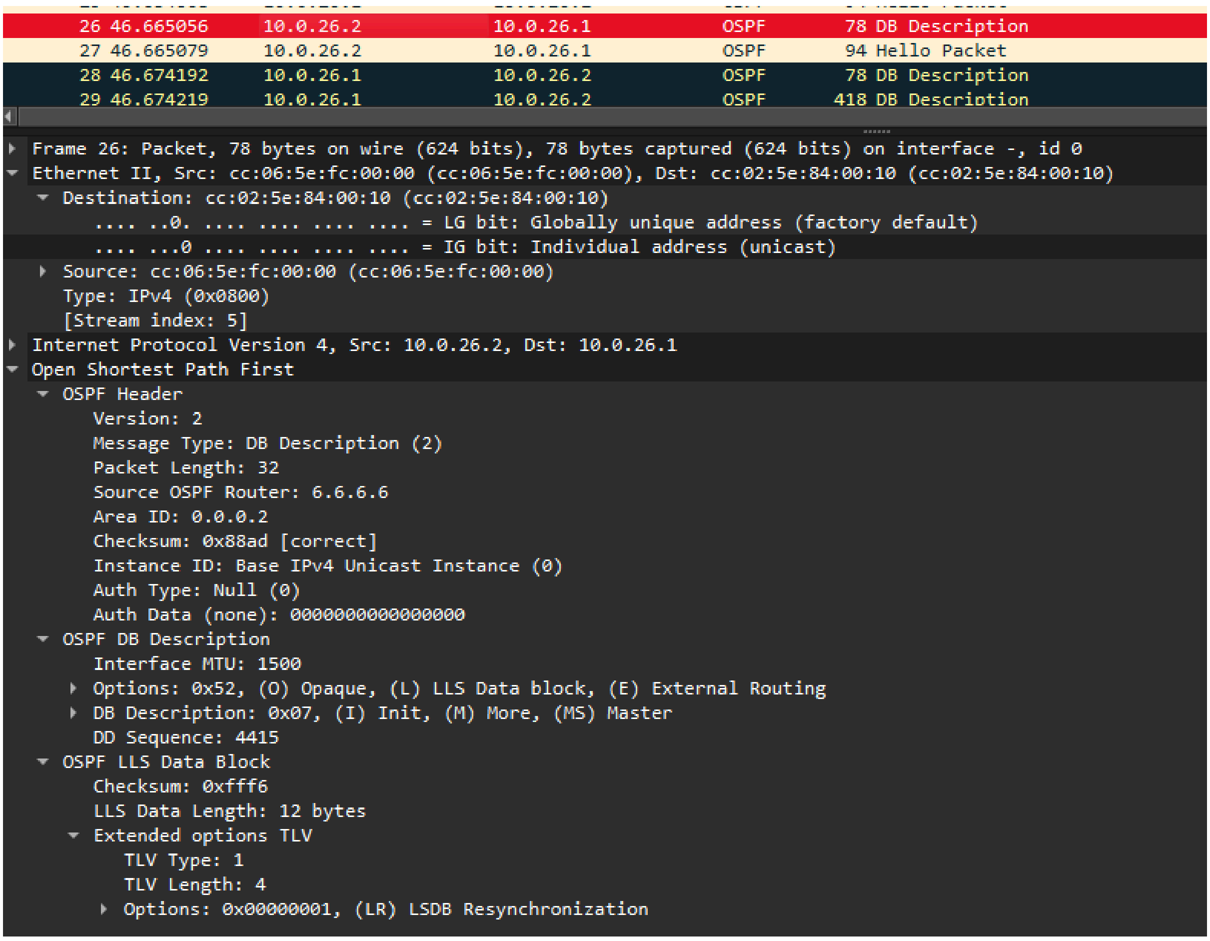
Message Type: Hello Packet
Source IP: 10.0.26.2
Destination IP: 224.0.0.5
Source OSPF Router: 6.6.6.6
Area ID: 0.0.0.2
Hello используется для обнаружения соседей и поддержания соседских отношений между маршрутизаторами OSPF.

```
22 46.604639   10.0.26.2   224.0.0.5   OSPF   122 LS Update
23 46.604680   10.0.26.2   224.0.0.5   OSPF    94 LS Update
24 46.644899   10.0.26.2   224.0.0.5   OSPF    90 Hello Packet
25 46.654068   10.0.26.1   10.0.26.2   OSPF    94 Hello Packet
26 46.665056   10.0.26.2   10.0.26.1   OSPF    78 DB Description
27 46.665079   10.0.26.2   10.0.26.1   OSPF    94 Hello Packet
28 46.674192   10.0.26.1   10.0.26.2   OSPF    78 DB Description
29 46.674219   10.0.26.1   10.0.26.2   OSPF   418 DB Description

Frame 22: Packet, 122 bytes on wire (976 bits), 122 bytes captured (976 bits) on interface -, id 0
Ethernet II, Src: cc:06:5e:fc:00:00 (cc:06:5e:fc:00:00), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
Internet Protocol Version 4, Src: 10.0.26.2, Dst: 224.0.0.5
Open Shortest Path First
  OSPF Header
    Version: 2
    Message Type: LS Update (4)
    Packet Length: 88
    Source OSPF Router: 6.6.6.6
    Area ID: 0.0.0.2
    Checksum: 0x3533 [correct]
    Instance ID: Base IPv4 Unicast Instance (0)
    Auth Type: Null (0)
    Auth Data (none): 0000000000000000
  LS Update Packet
    Number of LSAs: 1
    LSA-type 1 (Router-LSA), len 60
      .000 1110 0001 0000 = LS Age (seconds): 3600
      0... .... .... .... = Do Not Age Flag: 0
      Options: 0x22, (DC) Demand Circuits, (E) External Routing
      LS Type: Router-LSA (1)
      Link State ID: 6.6.6.6
      Advertising Router: 6.6.6.6
      Sequence Number: 0x80000004
      Checksum: 0x0c42
      Length: 60
      Flags: 0x00
      Number of Links: 3
      Type: Stub     ID: 192.168.63.0    Data: 255.255.255.0   Metric: 1
      Type: Transit  ID: 10.0.67.1       Data: 10.0.67.1       Metric: 1
      Type: Transit  ID: 10.0.26.1       Data: 10.0.26.2       Metric: 1
```

Message Type: LS Update
Source IP: 10.0.26.2
Destination IP: 224.0.0.5
Source OSPF Router: 6.6.6.6
Area ID: 0.0.0.2
Number of LSAs: 1
LSA-type 1: Router-LSA
Advertising Router: 6.6.6.6
Router-LSA описывает связи конкретного маршрутизатора внутри зоны. В данном случае маршрутизатор R6 сообщает о своих подключениях.
Внутри Router-LSA были указаны связи:
Stub network: 192.168.63.0
Transit network: 10.0.67.1
Transit network: 10.0.26.1
это значит, что маршрутизатор R6 объявляет свою пользовательскую сеть 192.168.63.0/24, а также связи с другими OSPF-маршрутизаторами через сети 10.0.67.0 и 10.0.26.0.



LSA-type 2: Network-LSA
Link State ID: 10.0.67.1
Advertising Router: 6.6.6.6
Netmask: 255.255.255.252
Attached Router: 6.6.6.6
Attached Router: 7.7.7.7
Network-LSA описывает транзитную сеть, в которой участвуют несколько OSPF-маршрутизаторов.
В данном случае описывается сеть между R6 и R7: 10.0.67.0/30
В поле Attached Router указаны
R6 — 6.6.6.6
R7 — 7.7.7.7



Message Type: DB Description
Source IP: 10.0.26.2
Destination IP: 10.0.26.1
Source OSPF Router: 6.6.6.6
Area ID: 0.0.0.2
Database Description используются при установлении OSPF-соседства, и передают краткое описание базы данных состояния каналов, то есть LSDB.
видны флаги
Init
More
Master

```
   35 49.103599      10.0.26.1             224.0.0.5           OSPF       138 LS Acknowledge

▾ OSPF Header
     Version: 2
     Message Type: LS Acknowledge (5)
     Packet Length: 104
     Source OSPF Router: 2.2.2.2
     Area ID: 0.0.0.2
     Checksum: 0x16f0 [correct]
     Instance ID: Base IPv4 Unicast Instance (0)
     Auth Type: Null (0)
     Auth Data (none): 0000000000000000
▾ LSA-type 1 (Router-LSA), len 60
     .000 1110 0001 0000 = LS Age (seconds): 3600
     0... .... .... .... = Do Not Age Flag: 0
   ▸ Options: 0x22, (DC) Demand Circuits, (E) External Routing
     LS Type: Router-LSA (1)
     Link State ID: 6.6.6.6
     Advertising Router: 6.6.6.6
     Sequence Number: 0x80000004
     Checksum: 0x0c42
     Length: 60
▾ LSA-type 2 (Network-LSA), len 32
     .000 1110 0001 0000 = LS Age (seconds): 3600
     0... .... .... .... = Do Not Age Flag: 0
   ▸ Options: 0x22, (DC) Demand Circuits, (E) External Routing
     LS Type: Network-LSA (2)
     Link State ID: 10.0.67.1
     Advertising Router: 6.6.6.6
     Sequence Number: 0x80000001
     Checksum: 0x940e
     Length: 32
▾ LSA-type 1 (Router-LSA), len 60
     .000 0000 0000 0010 = LS Age (seconds): 2
     0... .... .... .... = Do Not Age Flag: 0
   ▸ Options: 0x22, (DC) Demand Circuits, (E) External Routing
     LS Type: Router-LSA (1)
     Link State ID: 7.7.7.7
     Advertising Router: 7.7.7.7
     Sequence Number: 0x80000004
     Checksum: 0x58ca
     Length: 60
▾ LSA-type 2 (Network-LSA), len 32
     .000 0000 0000 0010 = LS Age (seconds): 2
     0... .... .... .... = Do Not Age Flag: 0
   ▸ Options: 0x22, (DC) Demand Circuits, (E) External Routing
     LS Type: Network-LSA (2)
     Link State ID: 10.0.67.2
     Advertising Router: 7.7.7.7
     Sequence Number: 0x80000001
     Checksum: 0x5c41
     Length: 32
```

Message Type: LS Acknowledge
Source IP: 10.0.26.1
Destination IP: 224.0.0.5
Source OSPF Router: 2.2.2.2
Area ID: 0.0.0.2

LS Acknowledge используется для подтверждения получения LSA.
Внутри пакета перечислены LSA, получение которых подтверждается, например:
Router-LSA от 6.6.6.6
Network-LSA для 10.0.67.1
Router-LSA от 7.7.7.7
Network-LSA для 10.0.67.2

7. Для PC show ip Для роутеров show ip route и show running-config